Homework 1 杨皓匡 未央-231

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1. suppose $x_2' = -x_2$, $s_1, s_2 \geq 0$ (slack variable)

the original question can be transformed into:

$$\text{maximize} \quad -8x_1 - 6x_2' - 5x_3$$

$$\text{s.t} \quad 2x_1 - 3x_2' \leq 5$$

$$x_1 + x_3 - s_1 \leq 2$$

$$x_1 - x_2' + x_3 - s_2 \leq 4$$

$$x_1, x_2', s_1, s_2, x_3 \geq 0.$$

2.

Suppose the number of full-time workers in January is x_1, so we suppose the number of advisers is x_1'

We believe that workers don't participate in production during training.

As a result, the constraint condition is:

$x_1 + x_1' \leq 50$ ①

$160x_1 + 110x_1' \geq 6000$ ②

$(50 \cdot 0.95 + x_1') - x_2') \cdot 160 + 110x_2' \geq 7000$ ③

$((50 \cdot 0.95 + x_1') \cdot 0.95 + x_2' - x_3') \cdot 160 + 110x_3' \geq 8000$ ④

$(((50 \cdot 0.95 + x_1') \cdot 0.95 + x_2') \cdot 0.95 + x_3' - x_4') \cdot 160 + 110x_4' \geq 9500$ ⑤

$((((50 \cdot 0.95 + x_1') \cdot 0.95 + x_2') \cdot 0.95 + x_4') \cdot 0.95 - x_5') \cdot 160$

$+ 110 \cdot x_5' \geq 11000$ ⑥

$x_1, x_1', x_2', x_3', x_4', x_5' \geq 0.$ ⑦

After simplified caculation:

① : $x_1 + x_1' \le 50$

② : $160x_1 + 110x_1' \ge 6000$

③ : $160x_1' - 50x_2' \ge -600$

④ : $152x_1' + 160x_2' - 50x_3' \ge 780$

⑤ : $144.4x_1' + 152x_2' + 160x_3' - 50x_4' \ge 2641$

⑥ $137.18x_1' + 144.4x_2' + 152x_3' + 160x_4' - 50x_5' \ge 4483.95$

⑦ : $x_1, x_1', x_2', x_3', x_4', x_5' \ge 0$.

At the same time, the number of workers to be paid each month is:

January : 50

February : $50 \cdot 0.95 + x_1'$

March : $(50 \cdot 0.95 + x_1') \cdot 0.95 + x_2'$

April : $((50 \cdot 0.95 + x_1') \cdot 0.95 + x_2') \cdot 0.95 + x_3'$

May : $(((50 \cdot 0.95 + x_1') \cdot 0.95 + x_2') \cdot 0.95 + x_3') \cdot 0.95 + x_4'$

After simplified caculation, the hole number of the cost which the factory should pay is:

$509556.9688 + 9048.7625x_1' + 7419.75x_2' + 5705x_3' + 3900x_4'$

(We suppose that the interns are unpaid).

As a result, the LP is:

minimal

$$509556.9688 + 9048.7625x_1' + 7419.75x_2' + 5705x_3' + 3900x_4'$$

$$\begin{aligned}
\text{s.t. } & x_1 + x_1' \le 50 \\
& 160x_1 + 110x_1' \ge 6000 \\
& 160x_1' - 50x_2' \ge -600 \\
& 152x_1' + 160x_2' - 50x_3' \ge 780 \\
& 144.4x_1' + 152x_2' + 160x_3' - 50x_4' \ge 2641 \\
& 137.18x_1' + 144.4x_2' + 152x_3' + 160x_4' - 50x_5' \ge 4483.95 \\
& x_1, x_1', x_2', x_3', x_4', x_5' \ge 0.
\end{aligned}$$

(We suppose the training for new workers will continue in May. if not, results will change)